

WANDO, Korea, Republic of

WMO: 471700

Lat: 34.3958N

Lon: 126.7019E

Elev: 37

StdP: 100.89

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.0	-2.7	-13.2	1.2	0.6	-11.2	1.4	0.5	13.3	1.2	12.3	1.0	4.7	340	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.7	31.7	26.4	30.5	25.9	29.4	25.3	27.3	30.1	26.6	29.5	25.9	28.8	3.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.6	22.2	29.1	25.7	21.0	28.5	25.0	20.2	27.9	86.9	30.3	83.4	29.7	80.5	28.9	31.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.6	10.0	8.7	-5.7	33.6	1.8	1.8	-7.0	34.9	-8.0	35.9	-9.0	36.9	-10.3	38.2		
			-7.1	27.9	1.5	1.3	-8.2	28.8	-9.1	29.6	-9.9	30.3	-11.0	31.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	2.7	4.1	7.9	13.1	17.9	21.6	25.2	26.4	22.6	17.3	11.0	5.0	(6)
(7)		DBStd	8.56	3.01	3.28	3.18	2.80	2.32	1.78	2.38	2.08	2.28	2.75	3.48	3.23	(7)
(8)		HDD10.0	667	227	167	79	6	0	0	0	0	1	29	158		(8)
(9)		HDD18.3	2085	485	399	322	156	36	1	0	1	51	221	414		(9)
(10)		CDD10.0	2360	1	2	16	101	246	347	472	508	379	228	59	2	(10)
(11)		CDD18.3	736	0	0	0	1	23	98	213	250	129	20	1	0	(11)
(12)		CDH23.3	4907	0	0	0	2	94	333	1621	2226	594	37	0	0	(12)
(13)		CDH26.7	1223	0	0	0	0	8	33	426	663	92	1	0	0	(13)
(14)	Wind	WSAvg	3.5	4.2	4.1	4.0	3.8	3.4	3.0	3.2	3.1	2.8	3.0	3.4	3.8	(14)
(15)	Precipitation	PrecAvg	1480	32	53	96	132	135	189	259	259	163	60	66	38	(15)
(16)		PrecMax	2258	57	148	180	284	268	441	565	620	545	186	240	87	(16)
(17)		PrecMin	894	4	6	21	24	47	25	41	31	22	4	3	7	(17)
(18)		PrecStd	324	16	32	44	66	54	104	126	147	139	42	55	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	15.7	18.8	23.2	27.7	29.1	33.0	33.8	30.5	26.0	21.1	14.9	(19)
(20)		0.4%	MCWB	9.5	12.8	12.4	14.8	17.9	22.0	26.9	26.9	24.8	20.1	17.1	11.4	(20)
(21)		2%	DB	10.1	12.9	16.6	21.3	25.7	27.6	31.5	32.1	28.8	24.3	19.1	12.7	(21)
(22)		2%	MCWB	7.0	9.7	11.5	14.6	17.3	21.0	26.3	26.6	23.5	18.5	15.3	9.4	(22)
(23)		5%	DB	8.6	11.0	14.8	19.7	24.1	26.5	30.3	31.0	27.6	23.1	17.6	11.2	(23)
(24)		5%	MCWB	5.8	7.3	10.5	13.9	16.7	20.9	26.0	26.3	22.7	18.2	14.2	8.3	(24)
(25)		10%	DB	7.1	9.2	13.3	18.2	22.6	25.4	29.1	30.0	26.5	21.7	16.1	9.8	(25)
(26)		10%	MCWB	4.3	5.9	9.5	13.2	16.4	20.7	25.5	25.7	22.3	17.1	13.1	7.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	14.1	15.0	18.9	21.6	24.7	28.8	28.6	26.1	23.1	18.2	12.5	(27)
(28)		0.4%	MCDWB	12.0	15.2	16.8	20.8	23.2	26.6	30.6	31.4	29.1	24.3	19.8	14.0	(28)
(29)		2%	WB	7.8	10.7	13.1	17.1	19.7	23.5	27.3	27.5	25.1	20.7	16.5	10.4	(29)
(30)		2%	MCDWB	9.5	12.3	15.0	19.1	22.6	25.5	29.7	30.6	27.5	22.6	18.2	12.2	(30)
(31)		5%	WB	6.2	8.0	11.5	15.6	18.7	22.6	26.4	26.8	24.2	19.2	15.1	8.8	(31)
(32)		5%	MCDWB	8.1	10.3	13.9	18.0	21.9	24.7	29.4	29.9	26.4	21.5	16.9	10.7	(32)
(33)		10%	WB	4.7	6.4	10.0	14.3	17.8	21.8	25.7	26.2	23.3	18.0	13.8	7.3	(33)
(34)		10%	MCDWB	6.7	8.6	12.8	16.7	20.9	24.0	28.6	29.2	25.4	20.9	15.6	9.6	(34)
(35)	Mean Daily Temperature Range		MDBR	5.7	6.9	7.9	8.2	8.0	6.4	5.4	5.7	6.5	7.4	7.0	6.2	(35)
(36)		5% DB	MCDBR	7.9	9.6	10.4	10.7	10.7	8.9	7.2	7.3	7.8	8.7	8.4	8.3	(36)
(37)			MCWBR	5.7	6.7	6.8	6.1	5.5	4.4	3.5	3.3	3.7	4.5	5.4	6.0	(37)
(38)			MCDBR	7.3	8.5	8.8	8.1	8.2	5.9	6.5	6.4	6.1	6.9	6.9	7.4	(38)
(39)		5% WB	MCWBR	6.1	7.0	7.0	6.3	5.1	3.7	3.8	3.3	3.6	4.7	5.5	6.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.448	0.491	0.554	0.568	0.583	0.631	0.599	0.552	0.501	0.469	0.456	0.444	(40)	
(41)		taud	1.968	1.896	1.768	1.789	1.806	1.782	1.912	2.040	2.115	2.067	2.024	1.987	(41)	
(42)		Ebn at Noon	716	733	723	740	734	698	718	745	764	748	704	690	(42)	
(43)		Edh at Noon	144	172	212	217	216	221	193	167	148	143	135	134	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.44	3.21	4.34	5.05	5.45	4.76	4.67	4.85	4.20	3.75	2.69	2.20	(44)	
(45)		RadStd	0.20	0.29	0.37	0.41	0.62	0.48	0.65	0.61	0.42	0.35	0.27	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page